Twitter Sentiment Analyzer

From Unsupervised **KMeans Clustering** to Supervised **Logistic Regression Classification**

Presented by :
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Project Overview

The Goal

To build a robust pipeline that categorizes tweets into four sentiments:

- Positive
- Negative
- Neutral
- Irrelevant

The Problem

Raw social media data is extremely noisy and unstructured.

Tweets often contain:

#Hashtags @Mentions

Emojis 😂 URLs

The Solution

A modular **Machine Learning pipeline** utilizing NLP techniques.

- Clean noisy text data
- Transform words to vectors
- Supervised Classification

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Project Structure (Modular Design)

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src/components

data_ingestion

Handles CSV loading and initial null cleaning from raw source.

data_transformation

Uses Lemmatization and TF-IDF to turn words into numerical vectors.

model_trainer

The "brain" where the Logistic Regression model is trained.

src/pipeline

train_pipeline

Connects all components for a one-click training process.

predict_pipeline

Loads saved artifacts to predict new, unseen tweets.

artifacts/

Stores generated **.pkl** files:
(Model, Vectorizer, Label Mapping)

ARCHITECTURE NOTE

This modular design ensures separation of concerns. The **Pipeline** orchestrates the flow, while **Components** handle specific logic, making the system scalable.

Why NOT Unsupervised KMeans?

How it works

KMeans attempts to find "groups" of similar words mathematically without actually knowing what the words mean or their sentiment context.

THE PITFALLS

Logic Gap

Words may be grouped by **topic** (e.g., "Xbox", "Game") rather than **sentiment**.

Poor Accuracy

Achieved only ~30% accuracy.

The model was essentially "guessing" the labels without ground truth.

Ambiguity

Often marks "Great service" as Negative just because it shares vocabulary with a Negative cluster.

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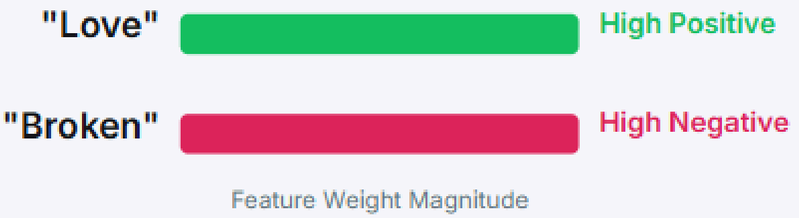
Why Logistic Regression is Better

Supervised Learning

Unlike KMeans which "guesses" groups, Logistic Regression uses actual **ground truth labels** (Positive/Negative) to learn patterns directly.

Mathematical Precision

It assigns specific numerical **"weights"** (coefficients) to every word in the vocabulary.



Performance Comparison

+50%
ACCURACY

Accuracy jumped to **80%+**

The model now understands the "intent" behind the words.

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The NLP Transformation Pipeline

1 Cleaning

Removing noise to standardize text.

@mentions URLs Punctuation

2 Stopwords

Removing common filler words that carry no sentiment value.

`"the", "is", "at", "and"`

3 Lemmatization

Reducing words to their base or dictionary root form.

`"running" → "run"`

4 TF-IDF Vector

Converting text to numbers based on word importance frequency.

`[0.42, 0.0, 0.15]`

PIPELINE IN ACTION

RAW TWEET

`"@Support I am loving this!"`

→

CLEANED

`"I am loving this"`

→

LEMMATIZED

`"love"`

→

VECTORIZED (Model Input)

`[0.0, 0.0, ... 0.75, ... 0.0]`

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Conclusion & Future Scope

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PROJECT OUTCOMES

Model Success

Successfully transitioned from guessing (KMeans) to a **Supervised Model**, achieving high-reliability results and clear intent classification.

Key Learning

Data Quality & **Model Selection** proved to be more important than complex algorithms alone.

FUTURE ROADMAP

Advanced NLP (BERT/Transformers)

Moving beyond simple word counting (TF-IDF) to deep **Context Understanding**. This will help detect sarcasm and complex sentence structures.

Real-time API Integration

Connecting directly to the **Twitter/X API** for live sentiment monitoring of trending topics instead of relying on static CSV datasets.

Thank you for your attention

User Interface (Streamlit)

An interactive dashboard built with **Streamlit** for a modern, responsive experience.

Single Predict

Real-time sentiment analysis of a manually typed tweet.

ENTER TWEET

Loving the new features! 🚀

Analyze

POSITIVE (98%)

Batch Predict

Upload a CSV and analyze thousands of tweets instantly.

Drag and drop file here

processing_tweets.csv 85%